

The digital human evolution

IF YOU'VE BEEN KEEPING ABREAST OF THE METAVERSE DEVELOPMENTS HAPPENING ALL ACROSS THE GLOBE, you've probably found yourself inundated with more information than you can ever possibly read or watch. The cacophony of the very large band on the wagon is reminiscent of the chaotic circus bands of the previous century that originated the phrase 'bandwagon' itself.

One aspect of that noise is about AI and how it will build the metaverse of the future. And of course it will. It makes total sense for a computer to be set to work designing and building fancy environments with little or no human supervision. That's the easy bit, and something already happening in gaming – the procedural generation of worlds. Such an AI would take a lot of examples and rote learn and then use permutations and combinations to make new worlds. They're not Turing Test capable, but they're certainly smart enough to be called AI. For those who don't know, a Turing test measures an AI's ability to fool humans into believing it is a real person.

Then there's the specialised AI which will help run the metaverse, reduce loads, parse data smartly, and do the myriad tasks that will need to be done in order to place millions of people simultaneously into a world. A simple thing like spatial audio in a Times Square (New York) equivalent inside the metaverse can be maddeningly complex. Just loads of people walking by you, talking as they move. In the real world you would drown out the hundreds of conversations as noise and focus on the one you're having, but still someone walking close to you would be heard for a split second. Imagine trying to code that into the metaverse? AI software, probably running on your computer as well as in the cloud, will decide in real-time how to adjust the volumes of hundreds of sounds to make it feel realistic. And that's just audio. The same has to be done for a million avatars all interacting with one another.

All of this assumes that we want the metaverse to mimic the real world. There's a good chance we won't!

Currently, Decentraland and Sandbox and other metaverses are making some of the same mistakes that Second Life did. They're very focussed on making a digital replica of the real world – with the quirks of the internet thrown in

of course. Like Second Life, the focus is on real estate and building things, but there's a reason Las Vegas-like stories don't happen too often. The "if you build it, they will come" idiom means nothing on the internet. If the internet was a real place, it would have an overpowering stench of rotting dead carcasses of products and services that believed in that motto and failed.

What's far more likely to be our metaverse of choice is exactly how we do multiplayer gaming online. We meet with friends and strangers alike, and play in much smaller groups than we'd normally see at a busy train station.

Our metaverse experience might end up with us interacting with AI most of the time and humans only on occasion. Even now, most of us stare at our phones whilst consuming content for a lot longer than we do at any other real human's face. Why do we think it will be any different in the metaverse? We will interact with avatars of friends, but more so with digital humans.

What is a digital human? We'd like to believe it's a human-looking avatar that's controlled by an AI, and which would pass the Turing Test. That might take a few more years or even decades to evolve, but we may not care. We already talk to Alexa, Siri and Google as if they are alive, even though no one sane will ever mistake them for a human. Now imagine Siri or Alexa as a real avatar, talking to you in the metaverse, moving around, with facial expressions and the like. Always there, available on demand, and existing to help you do anything you want in the metaverse. Ask them for directions, tell them to bring you stuff or take you somewhere, make them build you a house, bake you a fake cake, help you shop in the metaverse and have stuff delivered in the real world. These digital humans aren't a fantasy, they already exist. Companies have been developing them for customer service purposes for a long time. They started as chatbots, but they will end up as digital humans.

What's more, they will learn, adapt, evolve and grow with us as we find our own way through the metaverse. As we spend more and more time in the metaverse, digital humans will get better at mimicking and responding to us. While everyone is so focussed on how AI will help build the metaverse, I'm more intrigued by how we will help build the first AGI (artificial general intelligence). **d**



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